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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	SPARK SURGE
Project / Solution Name	LARIS: Quotation Tracker for Malaysian SMEs
Name (Leader)	HARRISH GOVIND A/L SETHULINGAM
Email & Phone (Leader)	harrishgovind09@gmail.com · [0125220471]
University Affiliation(s)	Universiti Tun Hussein Onn Malaysia (UTHM)
Live Prototype URL	https://laris-malaysia.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/lkaixian/laris

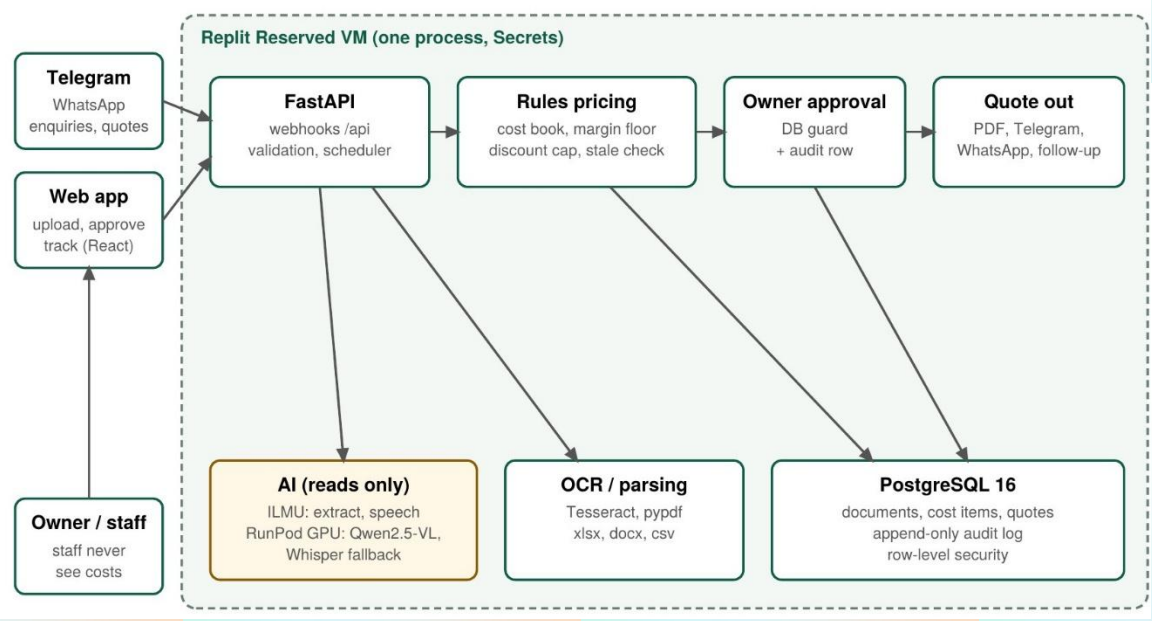
2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

LARIS runs as one FastAPI process on a Replit Reserved VM. It serves the React web app, the REST API (/api) and the Telegram and WhatsApp webhooks (/webhooks) from the same address.
Supplier quotes arrive by upload or chat. Text and tables are read directly (pypdf, openpyxl); scanned images go through Tesseract OCR; free text and voice notes go to the ILMU API, which returns items and quantities as JSON. Product photos go to Qwen2.5-VL on a RunPod Serverless GPU endpoint. Every model result is validated (confidence gate, arithmetic, dates, catalogue match) and anything uncertain is flagged for the owner.
Confirmed lines are written to Replit PostgreSQL as dated cost items. A rules engine in Python prices customer quotes from the cost book; models never set prices. The database enforces owner-only approval, an append-only audit log and row-level security between businesses. APScheduler runs follow-ups, expiry and data retention. Approved quotes go out as PDF or chat messages. Keys live only in Replit Secrets. The system block diagram is attached below.



2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.13	Backend language	PSF License
FastAPI 0.159.1 (Starlette 1.9.0)	API server, webhooks, serves the web app	MIT / BSD-3-Clause
uvicorn 0.34.0	ASGI web server	BSD-3-Clause
psycopg 3.3.6	PostgreSQL driver	LGPL-3.0
httpx 0.28.1	HTTP client for ILMU, RunPod, Telegram, WhatsApp, Mailtrap	BSD-3-Clause
APScheduler 3.11.3	Follow-ups, quote expiry, chat retention	MIT
pypdf 6.21.2	Text layer from supplier PDFs	BSD-3-Clause
Pillow 13.0.0	Image decoding and checks	MIT-CMU
pytesseract 0.3.13 + Tesseract OCR	OCR for scanned quotes	Apache-2.0
openpyxl 3.1.5, defusedxml 0.7.1	Read XLSX quotes, BOM export, safe XML parsing	MIT / PSF
reportlab 5.0.1	Customer quote PDF	BSD
cryptography 50.0.2	Ed25519 licence signature check	Apache-2.0 or BSD-3Clause
PyYAML 6.0.2	Trade templates	MIT
python-multipart 0.0.32	File uploads	Apache-2.0
TypeScript 7.0.2	Frontend language	Apache-2.0

Name & Version	Purpose in Project	License
React 19.3.1	Web user interface	MIT
Vite 8.3.5, @vitejs/plugin-react 6.1.2	Frontend build	MIT
Tailwind CSS 4.3.3	Styling	MIT
PostgreSQL 16	Database	PostgreSQL License
pytest 9.0.3, PGlite 0.5.11, axe-core 4.14.0	Automated tests, in-memory Postgres for tests, accessibility checks (development only)	MIT / Apache-2.0 / MPL-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
ILMU API (YTL AI Labs)	ilmu-mini-v3.3 extracts items and quantities from messages and quote text; ilmu-asr-v4.2 transcribes voice notes	Paid, pay per use
RunPod Serverless API (RunPod)	Calls Qwen2.5-VL-7B for product photos; Malaysian Whisper as speech-to-text fallback	Sponsor-provided credits
Telegram Bot API (Telegram)	Receive supplier and customer messages; send quotes and follow-ups	Free
WhatsApp Business Cloud API, Graph v21.0 (Meta)	Same as Telegram, primary customer channel	Free tier; paid per template message
Mailtrap Email API (Mailtrap)	Sign-up codes, password resets, invitations, quote emails	Free tier
NTP time servers (pool.ntp.org, time.google.com)	Trusted time for the software licence check	Free, public

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Reserved VM (Replit)	Hosts the single FastAPI process: web app, API, webhooks, scheduler	Reserved VM; region not stated by Replit
Replit PostgreSQL (Replit)	Main database: documents, cost book, quotes, jobs, audit log	Included with the Replit deployment

Service & Provider	Purpose in Project	Plan / Region
Replit Secrets (Replit)	Stores every API key and the database URL	Included
RunPod Serverless, vLLM worker, 24 GB GPU (RunPod)	Runs Qwen2.5-VL-7B and the Whisper fallback on demand	Serverless, pay per second; EU-SE-1
GitHub and GitHub Actions (GitHub)	Code repository and continuous integration (tests, audits)	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
ilmu-mini-v3.3	YTL AI Labs, ILMU API	Pre-trained, few-shot prompting, JSON output	Commercial API terms
ilmu-asr-v4.2	YTL AI Labs, ILMU API	Pre-trained	Commercial API terms
Qwen2.5-VL-7B-Instruct	Alibaba Qwen (Hugging Face), hosted on RunPod	Pre-trained	Apache-2.0
Mesolitica Malaysian Whisper	Mesolitica (Hugging Face), hosted on RunPod	Pre-trained, fallback only	Not stated on model card; base Whisper MIT
Tesseract 5 LSTM, English	tesseract-ocr project	Pre-trained	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
LARIS synthetic demo seed	https://github.com/lkaixian/laris (db/seed_made_to_order.sql)	Synthetic	Apache-2.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic supplier quotes and enquiries (8 documents, 3 enquiries)	https://github.com/lkaixian/laris (tests/fixtures/docs)	Synthetic	Apache-2.0	N
Trade templates	https://github.com/lkaixian/laris (config/templates)	Self-made	Apache-2.0	N
LARIS Demo Video Test Case	https://github.com/lkaixian/laris/demo_material	Synthetic	Apache-2.0	N
Published statistics, citation only (DOSM, UN SDG metadata, The Star)	dosm.gov.my, unstats.un.org, thestar.com.my	Public	Per source	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code with Claude Sonnet 5.5 models (Anthropic)	Code, tests, bug fixes, documentation, research notes, templates, pitch deck, demo video script	Major
Replit Repo AI Chat	Automation of git pull only.	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-written change runs through 506 automated tests and GitHub Actions CI before it is kept. The test cases were required to be written before the code generation happen. We clicked through every screen in a browser, re-ran the demo flow end to end, and fixed the bugs we found. Every figure in our documents traces to a cited source in the knowledge folder with a confidence label; unmeasured numbers are marked as not yet measured. The team reviewed the code and documents and takes responsibility for them.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Hosting platform (Reserved VM, PostgreSQL, Secrets)	Hosts the whole prototype: web app, API, webhooks, scheduler and database
RunPod	GPU cloud credits (Serverless)	Runs Qwen2.5-VL-7B for product photos and the Malaysian Whisper speech-to-text fallback

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential as the host of the running prototype, though the app is a standard FastAPI and PostgreSQL service that could move to another VM. RunPod is not essential to the core journey: supplier quote to cost book to priced quote to stale-cost alert runs without it. It adds photo reading and a speech fallback, and any OpenAI-compatible vision endpoint could replace it.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype holds synthetic data only. In real use it stores business users (name, username, email, hashed password), customer and supplier contacts, chat text, voice-note transcripts and supplier documents. Text, audio and photos are sent to ILMU, RunPod and Meta or Telegram to be read. Protection: scrypt password hashing, hashed session tokens, sign-in throttling, row-level security between businesses, staff cannot see costs, contact details masked before text reaches a model, keys only in Replit Secrets, append-only audit log. Retention: chat text is blanked after 180 days (RETENTION_DAYS), and an owner can erase a customer on request. Uploaded files are read in memory; only extracted lines are kept.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Models can misread blurred photos, thermal receipts or Manglish speech, and extraction accuracy on real Malaysian quotes is not yet measured. Mitigations: models never set prices; every line passes a confidence gate and arithmetic, date and catalogue checks; uncertain lines are flagged and the owner confirms before anything enters the cost book; every AI call is logged. Dependency on single providers (ILMU, RunPod, Meta) is softened by fallbacks: Whisper on RunPod for speech, Telegram for WhatsApp, and manual entry when no model is available. Rate limits and costs are not yet load-tested. The app now runs as one process, no failover.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

HARRISH.

24 OCTOBER 2026.